Chapter 6 The point estimation (点估计)

In general, we have two “constructive” methods for obtaining point estimators: the method of moments (矩法估计) and the method of maximum likelihood (极大似然估计).

The basic idea of the method of moments is to equate the kth sample moment , to the kth population moment. Then solve these equations to obtain the estimators.

Another example from the final exam

VII. (10 points) Let $X_1, X_2, \cdots, X_n$ be a random sample of size n from the pdf

$$f(x;\theta) = \begin{cases} \frac{x}{\theta} e^{-\frac{x^2}{2\theta}}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

(a) Use the method of moments to find an estimator for θ.

(b) Find the maximum likelihood estimator for θ.

$$E(X) = \int_{-\infty}^{+\infty} x f(x,\theta) dx = \int_{0}^{+\infty} \frac{x^2}{\theta} e^{-\frac{x^2}{2\theta}} dx = \frac{1}{2} \int_{-\infty}^{+\infty} \frac{x^2}{\theta} e^{-\frac{x^2}{2\theta}} dx$$

$$= \frac{\sqrt{2\pi}}{2\sqrt{\theta}} \int_{-\infty}^{+\infty} \frac{x^2}{\sqrt{2\pi}\sqrt{\theta}} e^{-\frac{x^2}{2\theta}} dx = \sqrt{\frac{\pi}{2\theta}} \cdot V(Y), \ (Y \sim N(0,\theta))$$

$$= \sqrt{\frac{\pi}{2\theta}} \cdot (\sqrt{\theta})^2 = \sqrt{\frac{\theta \cdot \pi}{2}}$$

$$\therefore \theta = \frac{2E^2(x)}{\pi}.$$

Thus, the moment estimator is $\hat{\theta}_M = \frac{2}{\pi} \overline{X}^2$.

方法2：$E(X)$求不出来，可以试试求$E(X^2)$

(or using 2rd moment)

$$E(X^2) = \int_{-\infty}^{+\infty} x^2 f(x,\theta)dx = \int_0^{+\infty} \frac{x^3}{\theta} e^{-\frac{x^2}{2\theta}} dx$$

$$= \int_0^{+\infty} x^2 e^{-\frac{x^2}{2\theta}} d\frac{x^2}{2\theta} = 2\theta \cdot \int_0^{+\infty} \frac{x^2}{2\theta} e^{-\frac{x^2}{2\theta}} d(\frac{x^2}{2\theta})$$

$$= 2\theta \cdot \int_0^{+\infty} u e^{-u} du = 2\theta$$

$$\therefore \theta = \frac{E(X^2)}{2}$$

The moment estimator is $\hat{\theta}_M = \frac{1}{2n} \sum_{i=1}^{n} X_i^2$.

More general case

Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with pmf or pdf $f(x; \theta_1, \ldots, \theta_m)$, where $\theta_1, \ldots, \theta_m$ are parameters whose values are unknown. Then the moment estimators $\hat{\theta}_1, \ldots, \hat{\theta}_m$ are obtained by equating the first m sample moments to the corresponding first m population moments and solving for $\theta_1, \ldots, \theta_m$.

Example (There are more than one parameters to be estimated)
Let $X_1, X_2, \dots, X_n$ be a random sample from a negative binomial distribution with parameters r and p. State the moment estimators of r and p (both r and p are unknown).

Observe that $E(X) = \frac{r}{p}$, $\qquad V(X) = r(1-p)/p^2$
and thus $E(X^2) = V(X) + [E(X)]^2 = r(\,1 - p + r)/p^2$

Equating $E(X) = \bar{X}$ and $E(X^2) = 1/n \sum X_i^2$, we obtain two equations with unknowns r and p. Solving these two equations yeilds the point estimators of r and p:

$$\hat{p} = \frac{\bar{X}}{\left(\frac{1}{n}\right)\sum X_i^2 - \bar{X}^2 + \bar{X}} \qquad \hat{r} = \frac{\bar{X}^2}{\left(\frac{1}{n}\right)\sum X_i^2 - \bar{X}^2 + \bar{X}}$$

r means the number of successes, and thus must be nonnegative. However, the moment estimators $\hat{r}$ may result in negative estimate of r. This indicates that the moment estimator is flawed.

Another "constructive" method for obtaining point estimators

Maximum Likelihood Estimation (极大似然估计)

Example

A sample of ten new bike helmets made by a certain company is obtained. Upon testing, it is found that the first, third, and tenth helmets are flawed, whereas the others are not. Let $p = P$(flawed helmet). Suppose that the conditions of various helmets are independent of one another. Find out a sensible estimate of p.

Intuitively, the p's value should be the one such that the observed sample is most likely to have occurred.

Question: how likely is the observed sample?

Let $X_1, X_2, \dots, X_{10}$ be the random sample.

$$X_i = \begin{cases} 1 & \text{the ith helmet is flawed} \\ 0 & \text{otherwise} \end{cases}$$

Then the observed $x_i's$ in this example are $1, 0, 1, 0, 0, 0, 0, 0, 0, 1$.

The joint pmf of the sample is

$$f(x_1, x_2, \dots x_{10}; p) = p(1-p)p \dots (1-p)p = p^3(1-p)^7.$$

Thus, the value of p is the one that maximizes the above pmf, or equivalently, maximizes the nature log of the above pmf:

$$\ln f(x_1, x_2, \dots x_{10}; p) = 3 \ln p + 7 \ln(1-p)$$

$$\frac{d}{dp}\{\ln f(x_1, x_2, \dots x_{10}; p)\} = \frac{d}{dp}\{3\ln p + 7\ln(1-p)\}$$

$$= \frac{3}{p} - \frac{7}{1-p}$$

Equating this derivative to 0 and solving for p gives $p = 0.3$.

In a word, when $p = 0.3$ the obtained sample is most likely to have occurred. Our point estimate is $\hat{p} = 0.3$

$\hat{p} = 0.3$ is called the maximum likelihood estimate because it is the parameter value that maximizes the likelihood (joint pmf) of the observed sample.

Definition

Let $X_1, X_2, \ldots, X_n$ have joint pmf or pdf

$$f(x_1, x_2, \ldots, x_n; \theta_1, \ldots, \theta_m),$$

where the parameters $\theta_1, \ldots, \theta_m$ have unknown values. When $x_1, x_2, \ldots, x_n$ are the observed sample values, the function f is regarded as a function of $\theta_1, \ldots, \theta_m$, which is called the likelihood function. The maximum likelihood estimates (mle's) $\hat{\theta}_1, \ldots, \hat{\theta}_m$ are those values of the θ_i's that maximize the likelihood function, so that

$$f\left(x_1, x_2, \ldots, x_n; \hat{\theta}_1, \ldots, \hat{\theta}_m\right) \geq f(x_1, x_2, \ldots, x_n; \theta_1, \ldots, \theta_m), \text{ for all } \theta_1, \ldots, \theta_m$$

when the X_i's are substituted in place of the x_i's, the maximum likelihood estimators are obtained.

Example

Suppose $X_1, X_2, \dots, X_n$ is a random sample from an exponential distribution with parameter λ. Because of independence, the likelihood function is a product of the individual pdf's:

$$f(x_1, x_2, \dots x_n; \lambda) = \left(\lambda e^{-\lambda x_1}\right) \cdots \left(\lambda e^{-\lambda x_n}\right) = \lambda^n e^{-\lambda \sum x_i}$$

The natural logarithm of the likelihood function is

$$\ln f(x_1, x_2, \dots x_n; \lambda) = n \ln \lambda - \lambda \sum x_i$$

Equating $(d/d\lambda)$[ln(likelihood)] to zero results in $\frac{n}{\lambda} - \sum x_i = 0$.

And thus, the mle is $\hat{\lambda} = n/\sum x_i$.

The maximum likelihood estimator of λ is $\hat{\lambda} = \frac{n}{\sum X_i} = 1/\bar{X}$

It is identical to the method of moment estimator.

Example

Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal distribution. The likelihood function is

$$f(x_1, \ldots, x_n; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x_1-\mu)^2/(2\sigma^2)} \cdot \cdots \cdot \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x_n-\mu)^2/(2\sigma^2)}$$

$$= \left(\frac{1}{2\pi\sigma^2}\right)^{n/2} e^{-\Sigma(x_i-\mu)^2/(2\sigma^2)}$$

so $$\ln[f(x_1, \ldots, x_n; \mu, \sigma^2)] = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\Sigma(x_i - \mu)^2$$

To find the maximizing values of μ and σ^2, we must take the partial derivatives of $\ln f$ with respect to μ and σ^2 and equate them to zero.

$$-\frac{1}{2\sigma^2}\sum 2(x_i-\mu)(-1)=\frac{\sum x_i - n\mu}{\sigma^2}=0$$

$$-\frac{n}{2}\left(\frac{1}{2\pi\sigma^2}\right)2\pi+\frac{1}{2\sigma^4}\sum(x_i-\mu)^2=-\frac{n}{2\sigma^2}+\frac{1}{2\sigma^4}\sum(x_i-\mu)^2=0$$

By the first equation $\hat{\mu}=\sum x_i/n$.

By the second equation $\widehat{\sigma^2}=\sum(x_i-\mu)^2/n$.

Then we substitute $\hat{\mu}=\sum x_i/n$ to obtain $\widehat{\sigma^2}=\frac{1}{n}\sum(x_i-\sum x_i/n)^2$.

And thus, the maximum likelihood estimator of μ and σ^2 are

$\hat{\mu}=\bar{X}$ and $\widehat{\sigma^2}=\frac{1}{n}\sum(X_i-\bar{X})^2$ respectively.

Estimating Functions of Parameters

The Invariance Principle (不变性原理)

Let $\hat{\theta}$ be the mle's of the parameters θ. Then the mle of any monotonic and invertible function $h(\theta)$ is the function $h(\hat{\theta})$.

Example

We have obtained the mle of σ^2 when the underlying distribution is normal. The mle of σ then is given by

$$\hat{\sigma} = \sqrt{\widehat{\sigma^2}} = \sqrt{\frac{1}{n}\sum(X_i - \bar{X})^2}$$

总体是离散分布，如何求参数的MLE?

七、（10 分）

(1) 设总体 X 等可能地取值1，2，3，…，N，其中 N 是未知的正整数。

$(X_1, X_2, \cdots, X_n)$是取自该总体中的一个样本．试求 N 的最大似然估计量．（7 分）

(2) 某单位的自行车棚内存放了 N 辆自行车，其编号分别为 1，2，3，…，N，假定职工从车棚中取出自行车是等可能的。某人连续 12 天记录下他观察到的取走的第一辆自行车的编号为

12， 203， 23， 7， 239， 45， 73， 189， 95， 112， 73， 159。
试求在上述样本观测值下，N 的最大似然估计值．（3 分）

Solution:

(1) The pmf of the population is $P(X = x) = \frac{1}{N}, (x = 1,2, \ldots, N)$.

The likelihood function is $f(x_1, x_2, \ldots x_n; N) = \frac{1}{N^n}, (1 \le x_i \le N)$.

Obviously, $\hat{N} = \max\{x_1, x_2, \ldots, x_n\}$.

The maximum likelihood estimator of N is $\hat{N} = \max\{X_1, X_2, \ldots, X_n\}$.

(2) The maximum likelihood estimate of N is

$$\hat{N} = \max\{x_1, x_2, \ldots, x_n\} = 239.$$

总体是连续分布，如何求参数的MLE?

七、(10 分)

设总体 X 的分布函数为

$$F(x;\beta)=\begin{cases}1-\dfrac{1}{x^{\beta}} & \text{当}x>1\\ 0 & \text{当}x\le 1\end{cases}$$

其中未知参数 $\beta>1, X_1, X_2, \cdots, X_n$ 为来自总体 X 的简单随机样本,求

(1) β 的矩估计;

(2) β 的极大似然估计。

(1) The method of moment

We firstly compute the first population moment $E(X)$.

Observe that the pdf of X is

$$f_X(x;\beta) = F'(x;\beta) = \begin{cases} \frac{\beta}{x^{\beta+1}}, & x > 1 \\ 0, & x \leq 1 \end{cases}.$$

And thus, $E(X) = \int_1^\infty \frac{\beta}{x^\beta} dx = \frac{\beta}{\beta-1}$.

Equating $E(X)$ to $\bar{X}$, we obtain $\hat{\beta}_1 = \frac{\bar{X}}{\bar{X}-1}$

(2) The method of maximum likelihood.

The pdf of X is

$$f_X(x;\beta) = F'(x;\beta) = \begin{cases} \frac{\beta}{x^{\beta+1}}, & x > 1 \\ 0, & x \le 1 \end{cases}.$$

And thus, the likelihood function is

$$f(x_1, x_2, \ldots x_n; \beta) = \frac{\beta^n}{x_1^{\beta+1} \ldots x_n^{\beta+1}} = \frac{\beta^n}{(x_1 \cdots x_n)^{\beta+1}}, (x_i \ge 1).$$

The natural logarithm of the likelihood function is

$$\ln f(x_1, x_2, \ldots x_n; \beta) = n \ln \beta - (\beta + 1) \sum \ln x_i$$

Equating $(d/d\beta)$[ln(likelihood)] to zero results in $\frac{n}{\beta} - \sum \ln x_i = 0.$

And thus, the mle is $\widehat{\beta}_2 = n/\sum \ln x_i$.

The maximum likelihood estimator of β is $\widehat{\beta}_2 = n/\sum \ln X_i$

Obviously, in most problems, there will be more than one reasonable estimator for a given parameter of interest.

Question: Which estimator is better?

Mean Square Error

Suppose $\hat{\theta}$ is an estimator of a parameter θ, which is a function of the sample X_i's. For some sample data, $\hat{\theta}$ will yield a value larger than θ, while for other sample data, $\hat{\theta}$ will underestimate θ. If we write

$$\hat{\theta} = \theta + \text{error of estimation}$$

then a good estimator would be one resulting in small estimation errors, so that estimated values will be near the true value.

Define mean square error MSE$= E\left[\left(\hat{\theta} - \theta\right)^2\right]$. If the first estimator has smaller MSE than does the second, it is natural to say that the first estimator is the better one.

The Unbiased Estimator

The MSE of an estimator $\hat{\theta}$ generally depends on the value of the unknown parameter θ, and thus finding an estimator with the smallest MSE is typically impossible.

So usually, we would find the best estimator among those with $\mathrm{E}(\hat{\theta}) = \theta$.

Definition

A point estimator $\hat{\theta}$ is said to be an unbiased estimator of θ if $\mathrm{E}(\hat{\theta}) = \theta$.

Clearly, the MSE of an unbiased estimator $\hat{\theta}$ is $V(\hat{\theta})$.

The Unbiased Estimator for the Population Mean and Variance

Proposition Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with mean μ and standard deviation σ. Then the estimator $\bar{X}$ is unbiased for estimating μ and

$$\hat{\sigma}^2 = S^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}$$

is unbiased for estimating σ^2 . This is because

$$\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i^2 - 2\bar{X}X_i + \bar{X}^2)$$

$$= \frac{1}{n}\sum_{i=1}^{n} X_i^2 - 2\bar{X}^2 + \bar{X}^2$$

$$= \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \bar{X}^2.$$

$$E\left[\frac{1}{n}\sum_{i=1}^{n}(X_i-\bar{X})^2\right]=\frac{1}{n}\sum E\left(X_i^2\right)-E[\bar{X}^2]$$

$$=\frac{1}{n}\sum(\sigma^2+\mu^2)-\ [V(\bar{X})+E(\bar{X})^2]$$

$$=\sigma^2+\mu^2-\left(\frac{\sigma^2}{n}+\mu^2\right)\ \ =\frac{n-1}{n}\sigma^2.$$

Thus, $\frac{\sum_{i=1}^{n}(X_i-\bar{X})^2}{n}$ is not unbiased for estimating σ^2. The **bias** is

$$\frac{n-1}{n}\sigma^2-\sigma^2=-\frac{1}{n}\sigma^2.$$

Since the bias is negative, the estimator with divisor n tends to underestimate σ^2, and this is why the divisor $n-1$ is preferred in application given that

$$E\left[\frac{1}{n-1}\sum_{i=1}^{n}(X_i-\bar{X})^2\right]=\frac{n}{n-1}\,E\left[\frac{1}{n}\sum_{i=1}^{n}(X_i-\bar{X})^2\right]=\sigma^2 \text{ is unbiased.}$$

Example Suppose that X, the reaction time to a certain stimulus, has a uniform distribution on the interval from 0 to an unknown upper limit θ. It is desired to estimate on the basis of a random sample $X_1, X_2, \dots, X_n$ of reaction times.

Since X has a uniform distribution on the interval from 0 to θ, $E(X) = \frac{1}{2}\theta$.

By equating $E(X)$ to $\bar{X}$, we obtain the moment estimator $\hat{\theta}_1 = 2\bar{X}$.

$E(\hat{\theta}_1) = E(2\bar{X}) = 2E(\bar{X}) = 2E(X) = \theta$,and thus $\hat{\theta}_1$ is an unbiased estimator.

And the MSE of $\hat{\theta}_1$ is $\mathrm{V}(\hat{\theta}_1) = V(2\bar{X}) = 4V(\bar{X}) = \frac{4V(X)}{n} = \frac{\theta^2}{3n}$.

On the other hand, since θ is the largest possible time in the entire population of reaction times, consider the largest sample reaction time, $\hat{\theta}_2 = \max\{X_1, X_2, \ldots, X_n\}$, as a second estimator.

Obviously $\hat{\theta}_2$ is a biased estimator, since we always have $\hat{\theta}_2 \le \theta$.

It is not hard to prove that $E(\hat{\theta}_2) = \frac{n}{n+1} \cdot \theta$.

And thus it is easy to modify $\hat{\theta}_2$ to obtain another unbiased estimator

$$\hat{\theta}_3 = \frac{n+1}{n} \cdot \max\{X_1, X_2, \ldots, X_n\}.$$

The expected value of this estimator is

$$E(\hat{\theta}_3) = \frac{n+1}{n} \cdot E(\hat{\theta}_2) = \frac{n+1}{n} \cdot \frac{n}{n+1} \cdot \theta = \theta.$$

Principle of Unbiased Estimation

When choosing among several different estimators of θ, select one that is unbiased.

Principle of Minimum Variance Unbiased Estimation

Among all estimators of θ that are unbiased, choose the one that has minimum variance. The resulting $\hat{\theta}$ is called the minimum variance unbiased estimator (MVUE) of θ.

Proposition

Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal distribution with parameters μ and σ. Then the estimator $\hat{\mu} = \bar{X}$ is the MVUE for μ.

Whenever we are convinced that the population being sampled is normal, the theorem says that $\bar{X}$ should be used to estimate μ.

Reporting a Point Estimate: The Standard Error

Besides reporting the value of a point estimate, some indication of its precision should be given. The usual measure of precision is the standard error of the estimator used.

The standard error of an estimator is its standard deviation $\sigma_{\hat{\theta}} = \sqrt{V(\hat{\theta})}$.

If the standard error itself involves unknown parameters whose values can be estimated, substitution of these estimates into $\sigma_{\hat{\theta}}$ yields the estimated standard error of the estimator.

The estimated standard error can be denoted by $\hat{\sigma}_{\hat{\theta}}$.

Example

Let X be a Bernoulli rv with $P(X = 1) = p$. Now we aim to estimate the parameter p.

Observe that $E(X) = p$. If $X_1, X_2, \dots, X_n$ is a random sample, then $\bar{X}$ is an unbiased estimator of p.

Let $\hat{p} = \bar{X}$. The standard error of $\hat{p}$ is

$$\sigma_{\hat{p}} = \sqrt{V(\bar{X})} = \sqrt{\sigma_X^2/n} = \sqrt{p(1-p)/n}$$

Since p *is* unknown, we substitute $\hat{p}$ into $\sigma_{\hat{p}}$, yielding the estimated standard error $\hat{\sigma}_{\hat{\theta}} = \sqrt{\hat{p}(1-\hat{p})/n}$.

Confidence Interval (置信区间)

Suppose we want to estimate a parameter θ of a single population.

A point estimate reports a single sensible value for θ.

Based on different sample data, we may obtain different point estimates $\hat{\theta}$. In general, $\hat{\theta} \neq \theta$.

A point estimate, because it is a single number, by itself provides no information about the precision and reliability of estimation.

An alternative is to calculate and report an entire interval of plausible values—an interval estimate or confidence interval (CI).

A confidence interval is calculated by first selecting a confidence level, which is a measure of the degree of reliability of the interval.

The confidence level

Suppose the confidence level (置信水平) is given. Based on different sample data, we may obtain different confidence intervals.

A confidence level of 95% implies that 95% of all samples would give an interval that includes θ.

The most frequently used confidence levels are $95\%, 99\%,$ and 90%.

The higher the confidence level , the more strongly we believe θ lies within the interval.

The precision of an interval estimate and the width of the interval

Information about the precision of a confidence interval is conveyed by the width of the interval.

If the confidence level is high and the resulting interval is quite narrow, our knowledge about the value of the parameter is reasonably precise.

A very wide confidence interval, however, gives the message that there is a great deal of uncertainty concerning the value of the parameter.

CI for the population mean

Case 1: CI for μ of a normal distribution with known σ

Suppose that the parameter of interest is a population mean μ and that

1. The population distribution is normal
2. The value of the population standard deviation σ is known

In the following, we will introduce the basic concepts and properties of confidence intervals.

Let $X_1, X_2, \ldots, X_n$ be a random sample. Then for any n, $\bar{X}$ is normally distributed with mean μ and standard deviation $\sigma/\sqrt{n}$.

Let $Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$. Then $Z \sim N(0,1)$.

Observe that $P(-1.96 \le Z \le 1.96) = 0.95.$

It means that $P\left(-1.96 \le \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \le 1.96\right) = 0.95.$

Since $-1.96 \le \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \le 1.96$

$\Longleftrightarrow -1.96 \cdot \sigma/\sqrt{n} \le \bar{X} - \mu \le 1.96 \cdot \sigma/\sqrt{n}$

$\Longleftrightarrow -\bar{X} - 1.96 \cdot \sigma/\sqrt{n} \le -\mu \le -\bar{X} + 1.96 \cdot \sigma/\sqrt{n}$

$\Longleftrightarrow \bar{X} - 1.96 \cdot \sigma/\sqrt{n} \le \mu \le \bar{X} + 1.96 \cdot \sigma/\sqrt{n}$

The inequality in the bracket means that the true value of μ lies in the following random interval

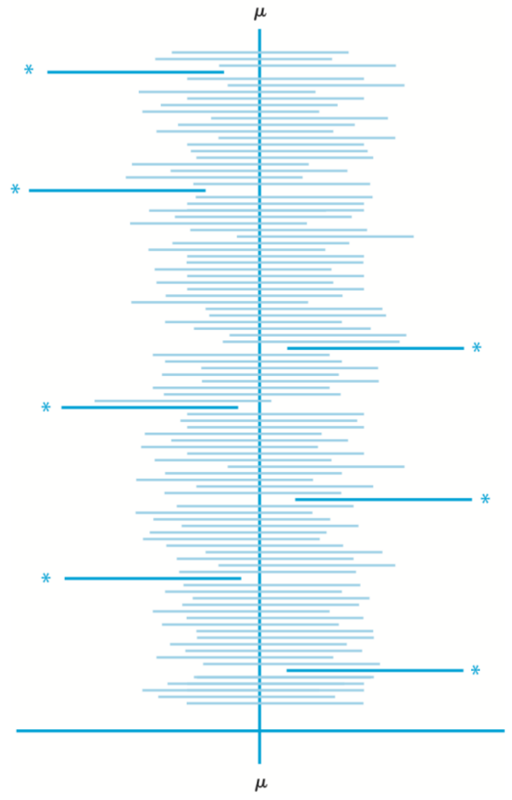
$$\left(\bar{X} - 1.96 \cdot \frac{\sigma}{\sqrt{n}}, \bar{X} + 1.96 \cdot \frac{\sigma}{\sqrt{n}}\right).$$

The random interval is centered at the sample mean $\bar{X}$ and extends $1.96 \cdot \sigma/\sqrt{n}$ to each side of $\bar{X}$.

Thus, the interval's width is $2 \cdot 1.96 \cdot \sigma/\sqrt{n}$, which is not random; only the location of the interval (its midpoint $\bar{X}$) is random.

The probability that the true value of μ lies in the random interval is 0.95. And thus, approximately 95 of the 100 computed intervals based on 100 different samples would contain μ.

Once a set of sample data is obtained, we substitute $\bar{x} = 80$ in place of $\bar{X}$, all randomness disappears. The resulting 95% confidence interval is a fixed interval.



One hundred 95% CIs (asterisks identify intervals that do not include μ)

Definition (95% confidence interval)

If after observing $X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n$, we compute the observed sample mean $\bar{x}$ and substitute $\bar{x}$ into $\left(\bar{X} - 1.96 \cdot \frac{\sigma}{\sqrt{n}},\ \ \bar{X} + 1.96 \cdot \frac{\sigma}{\sqrt{n}}\right)$ in replace of $\bar{X}$, the resulting fixed interval

$$\left(\bar{x} - 1.96 \cdot \frac{\sigma}{\sqrt{n}},\ \ \bar{x} + 1.96 \cdot \frac{\sigma}{\sqrt{n}}\right)$$

is called a 95% confidence interval for μ.

- A concise expression for the interval is $\bar{x} \pm 1.96 \cdot \frac{\sigma}{\sqrt{n}}$.

Example A sample of $n = 31$ trained typists was selected, and the preferred keyboard height was determined for each typist. The resulting sample average preferred height was $\bar{X} = 80$cm . Assuming that the preferred height is normally distributed with $\sigma = 2$cm , obtain the 95% CI for μ, the true average preferred height for the population of all experienced typists.

Solution The resulting interval is

$$\bar{x} \pm 1.96 \cdot \frac{\sigma}{\sqrt{n}} = 80 \pm 1.96 \times \frac{2}{\sqrt{31}} = 80 \pm 0.7 = (70.3, 80.7)$$

That is, we can be highly confident, at the 95% confidence level, that $70.3 < \mu < 80.7$. This interval is relatively narrow, indicating that μ has been rather precisely estimated.